

Punishment or Pruning? Diagnosing Algorithmic Collusion among Reinforcement-Learning Traders

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Abstract

Reinforcement-learning traders often settle on supracompetitive outcomes without communicating: in Kyle markets, Q-learning informed traders trade far less than competition implies, earning near-cartel profits and making prices less informative. Whether this is *collusion*, an outcome sustained by the threat of punishment, or a *learning bias* that prunes aggressive actions matters for policy: the first calls for limits on transparency, the second for standards on algorithm design. We develop diagnostics that separate the two: a paired rival-deviation test, a noise-shock test, a no-memory control, learning-rate sensitivity, and a new *myopic placebo* that retrains the same learners with no concern for the future and therefore no reason to punish. We show analytically that in the standard Kyle market a one-period deviation moves order flow by only $(I - 1)/(2I)$ noise standard deviations per unit of the value signal, whatever the volume of noise trading, while with a large mass of information-insensitive investors the ratio is in the hundreds, which explains why punishment-based collusion has been reported only in that regime. In simulations with hundreds of independent markets per configuration, Q-learners in the standard Kyle market reach up to – of the way from competition to the cartel, yet rivals never punish deviations, even with perfect monitoring of each other’s orders, and deviating always pays. We release a validated, CPU-only testbed with exact benchmarks, tabular and deep RL agents, and all diagnostics.

1 Introduction

Trading is increasingly delegated to algorithms that learn from their own profits. A growing literature shows that such algorithms can arrive at *supracompetitive* outcomes without communicating: reinforcement-learning (RL) pricing agents settle on prices above the competitive level [5, 18], algorithmic market makers quote wider spreads [8, 9], and RL informed speculators trade less aggressively than competition would imply, earning collusive profits and degrading price informativeness [12]. For regulators the finding is alarming, because tacit collusion by algorithms falls outside the reach of rules written around agreements among humans [15, 17].

A supracompetitive outcome, however, is not the same thing as collusion. In the game-theoretic sense, collusion is an outcome sustained by the threat of punishment: a trader who deviates to grab a larger share of the rent is punished by its rivals, so deviating does not pay. The same low-intensity outcome can instead come from a learning bias: an algorithm that systematically undervalues aggressive actions will trade conservatively even when no rival could ever punish it. Dou et al. [12] name both channels in the trading context (“price-trigger” collusion and “over-pruning” bias), and a parallel debate in industrial organization asks whether Q-learning collusion is genuine or an artifact of exploration and learning design [1, 2, 4, 6, 10]. The distinction is not academic. If algorithms punish, transparency matters, because monitoring is what makes punishment possible, and the

natural remedies target information: reporting delays, reduced order-flow disclosure. If the outcome is a learning bias, transparency is irrelevant and the remedies target algorithm design.

This paper asks how to tell the two apart, and what the answer is for RL informed traders in the canonical Kyle [19] market and in the Dou et al. [12] extension with information-insensitive investors.

Contributions.

1. **A diagnostic toolkit.** We combine five diagnostics that separate punishment from pruning: (i) a paired *rival-deviation* test that forces one trader to play its one-period best response and measures the rivals' reaction under common random numbers; (ii) the Dou et al. [12] *noise-shock* impulse response; (iii) a *myopic placebo*: the same learners with discount factor $\gamma = 0$, which have no reason to punish or to fear punishment, so any collusive signature they display is by construction not strategic; (iv) a *no-memory control* that removes the information punishment requires; and (v) *learning-rate sensitivity*, since a learning bias should move with the step size while a strategic equilibrium should not.
2. **A detectability result.** We show analytically that in the standard Kyle market a one-period deviation from the collusive intensity moves order flow by only $(I - 1)/(2I)$ noise standard deviations per unit of v/σ_v , whatever the volume of noise trading (Proposition 2). Deviations are close to invisible, so price-trigger strategies have little to work with. With a large mass of information-insensitive investors the ratio becomes $\xi\sigma_v(I - 1)/(4I\sigma_u)$, which is in the hundreds in the Dou et al. [12] calibration. This explains, from first principles, why punishment-based collusion has been reported only in that regime.
3. **Evidence.** In the standard Kyle market, tabular Q-learners reach up to – of the way from competition to the cartel outcome, yet rivals do not react to deviations, deviations pay, and the under-trading moves with the learning rate. The results are consistent with pruning, not punishment.
4. **An open testbed.** All experiments run on a batched simulator (hundreds of independent markets per process, CPU only) with exact Nash and cartel benchmarks, tabular Q-learning, DQN and PPO agents, checkpointing, and a test suite that validates the economics and the diagnostics against scripted traders whose behaviour is known.

Why this matters for policy. The two channels call for different interventions. Our results suggest that in markets like the standard Kyle setting, the supracompetitive outcomes produced by off-the-shelf Q-learning are properties of the learning procedure, and would respond to algorithm-design standards (step sizes, exploration) rather than to limits on transparency. Where deviations are highly detectable, as with a large mass of information-insensitive investors, the diagnostics must be applied with a placebo, because a reaction to a price shock is not by itself evidence of punishment.

2 Related Work

Algorithmic collusion. Calvano et al. [5] show that Q-learning agents in a repeated Bertrand game learn supracompetitive prices sustained by punishment, documented through impulse responses to a forced deviation; Klein [18] obtains similar results under sequential pricing. A second wave questions how these outcomes arise. Asker et al. [2, 3] show that the outcome depends on algorithm

design: asynchronous updating (learning only from the action taken) yields near-monopoly prices, while synchronous, counterfactual updating yields competitive ones. Abada and Lambin [1] attribute seemingly collusive outcomes to imperfect exploration and show that a regulator can restore competition by intervening in the learning process. Banchio and Mantegazza [4] show that collusion can arise from spontaneous coupling of the learning dynamics rather than from punishment strategies. den Boer et al. [10] argue that the collusive equilibria of Q-learning are reached only on time scales irrelevant in practice and require implausibly synchronized algorithms. Calvano et al. [6] discuss when algorithmic collusion is genuine or spurious. Deep RL agents [11] and large language models [16] can also price above competitive levels. We contribute a placebo-based diagnostic (the myopic learner) and a detectability analysis to this debate, in a financial market with adverse selection.

Algorithms in financial markets. Colliard et al. [8] study Q-learning market makers and find mark-ups over the competitive price driven by limited experimentation and noisy feedback, the same mechanism we find for informed traders. Cartea et al. [7] show that the tick size bounds the excess rents of learning liquidity providers, and Cont and Xiong [9] study tacit collusion among market-making algorithms in dealer markets. Dou et al. [12] extend Kyle [19] to several informed speculators, information-insensitive investors, and a market maker who trades off pricing error against inventory. They show that Q-learning speculators sustain collusive profits and reduce price informativeness, and separate price-trigger collusion from over-pruning bias with noise-shock impulse responses, finding the former only when information-insensitive investors are present. An independent reimplementation [13] reproduces both regimes and finds that the level of collusion depends on the update protocol. Our paper adds the rival-deviation test, the myopic placebo and the learning-rate sensitivity analysis, explains the regime difference analytically, and provides a validated open testbed.

Estimation bias in Q-learning. The max operator makes Q-learning over-estimate action values under noise [24, 25]. The bias relevant here runs the other way for rarely chosen actions: with a constant step size and ε -greedy exploration, an action whose estimate falls below the greedy action by chance is rarely updated again, so pessimistic errors persist while optimistic ones are corrected. Actions with noisier payoffs are more exposed. Decreasing step sizes that satisfy the Robbins and Monro [21] conditions remove the persistent component asymptotically [14].

3 Model

3.1 The repeated market

Time is discrete, $t = 1, 2, \dots$. Each period a fresh asset value v_t is drawn i.i.d. with mean $\bar{v}$ (normalised to 0) and variance σ_v^2 . $I \geq 2$ learning informed traders observe v_t and submit market orders $x_{i,t}$. Noise traders submit $u_t \sim \mathcal{N}(0, \sigma_u^2)$, and $P \geq 0$ optional *passive* informed traders play a fixed Nash strategy. Following Dou et al. [12], a mass $\xi \geq 0$ of information-insensitive investors demands $z_t = -\xi(p_t - \bar{v})$. A market maker observes aggregate informed and noise flow $y_t = \sum_i x_{i,t} + u_t$ and sets

$$p_t = \arg \min_p \theta (p - \mathbb{E}[v_t | y_t])^2 + (y_t + z_t)^2 \implies p_t = \bar{v} + \lambda y_t, \quad \lambda = \frac{\theta \lambda_B + \xi}{\theta + \xi^2}, \quad (1)$$

where $\lambda_B = \text{Cov}(v, y) / \text{Var}(y)$ is the Bayesian price impact. With $\xi = 0$ this is the Kyle [19] market maker. Trader i earns $\pi_{i,t} = (v_t - p_t) x_{i,t}$. The value is revealed at the end of the period, and the one-period game repeats.

3.2 Benchmarks

With linear strategies $x_i = \beta_i v$, profits depend only on second moments, so the benchmarks below hold for any value distribution as long as pricing is linear. Let $n = I + P$.

Proposition 1 (Nash and collusive benchmarks). *Let $\xi = 0$. The symmetric Nash equilibrium of the stage game has $\beta^N = \sigma_u/(\sqrt{n}\sigma_v)$ and $\lambda^N = \sqrt{n}\sigma_v/((n+1)\sigma_u)$. If the learners jointly maximise their profit against a market maker who prices their joint strategy rationally, while passive traders keep playing β^N (aggregate $B_P = P\beta^N$), the learners' aggregate intensity is*

$$B_L^C = \sqrt{B_P^2 + \sigma_u^2/\sigma_v^2}.$$

For $\xi > 0$, the Nash intensity solves $\beta = 1/((n+1)\lambda(n\beta))$, which has a unique root, and the collusive intensity maximises $B_L\sigma_v^2(1 - \lambda(B_L + B_P)(B_L + B_P))$.

Collusion lowers aggregate intensity ($B_L^C < I\beta^N$ for $P = 0$), raises learners' profit, and reduces price informativeness $\mathcal{I} = \text{Corr}(p, v)^2 = B^2\sigma_v^2/(B^2\sigma_v^2 + \sigma_u^2)$. Following Calvano et al. [5] we report any outcome metric m as a *collusion index*

$$\Delta_m = \frac{m - m^N}{m^C - m^N},$$

so $\Delta = 0$ is competition and $\Delta = 1$ is the cartel. We use aggregate intensity (the estimated slope of learners' total order on v), informativeness, and profit. Because the Nash–cartel profit gap is only 6% at $I = 2$, intensity and informativeness are the more precise measures.

Remark 1 (Passive traders can remove the gain from collusion). With $\xi = 0$ and $\sigma_v = \sigma_u$, $B_L^C = \sqrt{P^2/n + 1}$ and $I\beta^N = I/\sqrt{n}$, which coincide exactly when $I = P + 1$. In that case the learners have nothing to gain from colluding and Δ is undefined. We avoid this ratio when using passive traders as an intervention.

3.3 How detectable is a deviation?

Punishment requires monitoring. The relevant quantity is how much a deviation moves what traders observe relative to noise. Suppose rivals play the collusive intensity and the market maker prices at λ^C . A trader's myopic best response is $\beta^d = (1 - \lambda^C(I - 1)\beta^C)/(2\lambda^C)$, and a one-period deviation moves aggregate flow by $(\beta^d - \beta^C)v$ against noise of standard deviation σ_u .

Proposition 2 (Deviation signal-to-noise). *Let $P = 0$ and define $\text{SNR} = (\beta^d - \beta^C)\sigma_v/\sigma_u$, the flow change of a best-response deviation per unit of v/σ_v , in noise standard deviations.*

1. If $\xi = 0$, $\text{SNR} = (I - 1)/(2I) < 1/2$ for all I , independently of σ_u and σ_v .
2. As $\xi \rightarrow \infty$ with θ fixed, $\text{SNR} = \xi\sigma_v(I - 1)/(4I\sigma_u)(1 + o(1))$.

The proof is in Appendix A. In the standard Kyle market every intensity scales with σ_u/σ_v , so changing noise volume rescales deviations and noise alike and cannot make monitoring easier. A single deviation is less than half a noise standard deviation even at $|v| = \sigma_v$, so a trigger strategy tuned to catch it would mostly fire on noise. With information-insensitive investors, price impact is pinned near $1/\xi$, intensities scale with ξ , and deviations become enormous relative to noise: in the Dou et al. [12] calibration ($\sigma_v = 1$, $\sigma_u = 0.1$, $\xi = 500$, $\theta = 0.1$, $I = 2$) the ratio is about 625. Detectability therefore predicts exactly the regime split that Dou et al. [12] report: over-pruning only at $\xi = 0$, price triggers at $\xi = 500$.

4 Learning Traders and Diagnostics

4.1 Learning algorithms

Tabular Q-learning. Each trader in each market owns a Q-table indexed by a memory state s (what it remembers about the previous period), the current value v , and an order from a finite grid. It plays ε -greedy with $\varepsilon_t = e^{-\beta t}$ and updates

$$Q(s_t, v_t, x_t) \leftarrow Q(s_t, v_t, x_t) + \alpha_t \left[\pi_t + \gamma \max_{x'} Q(s_{t+1}, v_{t+1}, x') - Q(s_t, v_t, x_t) \right].$$

Following Calvano et al. [5] and Dou et al. [12], Q is initialised to the discounted payoff of each order against uniformly random rivals, and the step size is constant ($\alpha_t = \alpha$) unless stated otherwise. As an alternative we use per-entry decaying step sizes $\alpha_n = \alpha(1 + n/\kappa)^{-0.7}$, where n counts updates of the entry, which satisfy the Robbins and Monro [21] conditions.

Deep RL. Double DQN [20, 26] and PPO with generalised advantage estimation [22, 23] use the same information as continuous features (current value, previous value, and the memory signal), the same order grid, and one independent network per trader per market.

Memory. We vary what traders remember from the previous period: *none* (no memory; punishment is impossible); *flow* (binned aggregate order flow); *residual* (previous value and binned flow net of own order, i.e. what everyone else and noise traders did); *orders* (previous value and the rival’s exact order: perfect monitoring); and *price* (previous value and binned price surprise, the state used by Dou et al. [12]).

Market maker. The market maker re-estimates λ_B from exponentially weighted moments of (v, y) with a half-life of 2,000 periods and prices with (1), so it responds to the learners’ strategies as they evolve.

4.2 Diagnostics

After training, exploration and learning are switched off and greedy play is evaluated for 20,000 periods while the market maker keeps adapting.

D1: rival-deviation test. From a point on the greedy path we clone each market. In the clone, trader 1 plays its myopic best response for one period; afterwards everyone follows the learned strategies in both copies, which share every value and noise draw. We report the rivals’ change in trading intensity at each lag k , $\Delta\beta_k^{\text{rival}} = \mathbb{E}[v_k \Delta x_k^{\text{rival}}] / \text{Var}(v)$, and the deviator’s discounted gain over the window. Punishment means $\Delta\beta_1^{\text{rival}} > 0$ and a negative gain. This is the test Calvano et al. [5] use for pricing algorithms, adapted to a stochastic market with paired simulations.

D2: noise-shock test. Following Dou et al. [12], the clone receives an extra noise-trader order in the direction of v , moving the price the way a rival’s over-trading would although nobody deviated. Under price-trigger strategies traders trade harder next period; under over-pruning they do not react. We size shocks in *deviation units*: 1 is the flow change of a best-response deviation at $|v| = \mathbb{E}|v|$, so shocks mean the same thing in both regimes (Proposition 2).

D3: myopic placebo. We retrain the same learners with $\gamma = 0$. A myopic trader values only today’s profit, so it gains nothing from punishing and fears no punishment. Its Q-values estimate immediate expected profit, and any systematic reaction it shows to deviations or shocks cannot be a punishment strategy. If myopic learners reproduce a “collusive” outcome or signature, that outcome or signature does not identify collusion.

D4: no-memory control. With memory *none*, traders cannot condition on the past, so no punishment strategy exists; any supracompetitive outcome is a learning bias.

D5: learning-rate sensitivity. A strategic equilibrium is a property of the game; a bias of the learning procedure should scale with the step size and weaken under decaying step sizes. We vary the constant α and compare with decaying step sizes.

Convergence. Constant-step Q-learning with noisy payoffs keeps estimates moving, so greedy strategies need not settle. We report the share of greedy table entries that changed over the last 10^5 training periods and the mean on-path change in grid steps.

4.3 Experimental design and implementation

Unless noted, $I = 2$, $P = 0$, $\sigma_v = 1$, $\gamma = 0.95$, and each configuration is run for 100–200 independent markets. Standard-Kyle experiments ($\xi = 0$) use $\sigma_u = 1$, 5 values and a 31-point order grid spanning ± 1.5 times the largest Nash order; results in this regime are invariant to σ_u (Proposition 2). The Dou et al. [12] replication uses their calibration ($\sigma_u = 0.1$, $\theta = 0.1$, $\alpha = 0.05$), 10 values, a 15-point per-value grid spanning the cartel-to-Nash range plus 10%, and the price state. Confidence intervals are 95% across markets. The simulator advances all markets of a configuration as one batched array operation, which makes each experiment a matter of minutes to hours on a 4-core CPU. The code, configurations and a 70-test suite are released with the paper; the tests check the benchmarks against numerical optimisation, reproduce the Nash and cartel outcomes when the corresponding strategies are played, and verify that D1 and D2 report a reaction for scripted punishing traders and none for scripted memoryless ones.

5 Results

Results pending.

6 Discussion

Discussion pending.

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A Proofs

Proposition 1. With $x_j = \beta_j v$ and $p = \lambda y$, trader i ’s expected profit is $\beta_i \sigma_v^2 (1 - \lambda \sum_j \beta_j)$ (the noise term has zero mean). Taking λ as given, the first-order condition gives $\beta_i = (1 - \lambda \sum_{j \neq i} \beta_j) / (2\lambda)$, so a symmetric equilibrium has $\beta = 1 / ((n + 1)\lambda)$. With $\xi = 0$, rational pricing gives $\lambda = n\beta\sigma_v^2 / (n^2\beta^2\sigma_v^2 + \sigma_u^2)$. Substituting and solving yields $\beta^N = \sigma_u / (\sqrt{n}\sigma_v)$ and $\lambda^N = \sqrt{n}\sigma_v / ((n + 1)\sigma_u)$. For the collusive benchmark, total learner profit with rational pricing and aggregate $T = B_L + B_P$ is $B_L\sigma_v^2\sigma_u^2 / (T^2\sigma_v^2 + \sigma_u^2)$. The first-order condition in B_L is $T^2\sigma_v^2 + \sigma_u^2 - 2B_L T\sigma_v^2 = 0$, i.e. $\sigma_v^2(B_P^2 - B_L^2) + \sigma_u^2 = 0$, which gives B_L^C . For $\xi > 0$, $h(\beta) = (n + 1)\beta\lambda(n\beta) - 1$ is strictly increasing because both $\beta\lambda_B(n\beta) = n\beta^2\sigma_v^2 / (n^2\beta^2\sigma_v^2 + \sigma_u^2)$ and $\xi\beta$ are increasing, and $h(0) = -1$, $h(\infty) > 0$, so the root is unique. $\square$

Proposition 2. (i) With $\xi = 0$ and $P = 0$, $\beta^C = \sigma_u / (I\sigma_v)$ and $\lambda^C = \sigma_v / (2\sigma_u)$. Then $\beta^d = (1 - \lambda^C(I - 1)\beta^C) / (2\lambda^C) = \frac{\sigma_u}{\sigma_v} (1 - \frac{I-1}{2I}) = \frac{\sigma_u}{\sigma_v} \cdot \frac{I+1}{2I}$, so $\beta^d - \beta^C = \frac{\sigma_u}{\sigma_v} \cdot \frac{I-1}{2I}$ and $\text{SNR} = (I - 1) / (2I)$. (ii) From (1), $\lambda = 1/\xi + O(\xi^{-2})$ uniformly in $\lambda_B \in [0, \sigma_v / (2\sigma_u)]$. Treating λ as constant, the collusive aggregate maximises $B(1 - \lambda B)$, so $\beta^C = 1 / (2I\lambda)$, and $\beta^d = (1 - \lambda(I - 1)\beta^C) / (2\lambda) = \frac{1}{2\lambda} \cdot \frac{I+1}{2I}$. Hence $\beta^d - \beta^C = \frac{1}{\lambda} \cdot \frac{I-1}{4I} = \xi \frac{I-1}{4I} (1 + o(1))$ and $\text{SNR} = \xi\sigma_v(I - 1) / (4I\sigma_u)(1 + o(1))$. Numerically, the exact ratio at $I = 2$, $\sigma_v = 1$, $\sigma_u = 0.1$, $\theta = 0.1$ is 62.49, 625.00 and 6250.0 for $\xi = 50, 500, 5000$. $\square$